

Addis Ride demand Forecast

Short-term demand forecasting for on-demand rides using ML and spatial analytics



The problem: how many rides, where, and when?

Forecast hourly trips in 12 Addis Ababa zones for 1–14 November 2025 (4,032 zone-hours)

- **Why it matters:** drivers in the right zone at the right hour mean shorter waits and more completed trips

Trips (85k rows, Jan–Oct): hourly requests per zone, plus fares, wait times and active drivers

- messy: 3 date formats, 55 spellings of 12 zones, –1 codes, duplicates, an outage, a zone launched in March

Weather (hourly, observed + November forecast): temperature, rain, humidity, wind

- trap: recorded on UTC, not local time; –9999 rain codes; a July block in °F

Events (165): football, concerts, conferences, road closures, holidays

- messy: multi-zone and city-wide entries, missing or reversed end times, free-text crowd sizes, cancellations

- **Our goal:** one clean zone × hour table that joins all three, then a model that uses only what is known 14 days ahead

Cleaning & integration: one clock, one zone key, one row per zone-hour

- **The clock problem:** the weather file is on UTC. Raw temperature peaks at 12:00; after +3 h it peaks at 15:00 EAT, and rain–demand correlation jumps from 0.06 to 0.42
- **Join map:** zone-hour grid ← weather (many-to-one on the hour) ← events (interval join from 2 h before start to 2 h after end, per affected zone)
- **Key numbers (A4):**
 - 38,574 timestamps parsed explicitly (no day/month swaps)
 - 32,391 zone labels → 12 canonical zones
 - 846 duplicates, 675 -1 codes, 123 impossible spikes removed
 - 353 °F readings converted, 76 -9999 rain codes, 192 missing weather hours filled
 - 42-hour city-wide outage and Ayat before launch excluded, not zero-filled
 - Weather match 100%; joins never change row counts; 13/13 integrity checks pass
- **Result:** master_train has 82,979 zone-hours, master_test has 4,032

fig06 – Weather timestamps are UTC: shifting +3 h aligns temperature and rain with demand

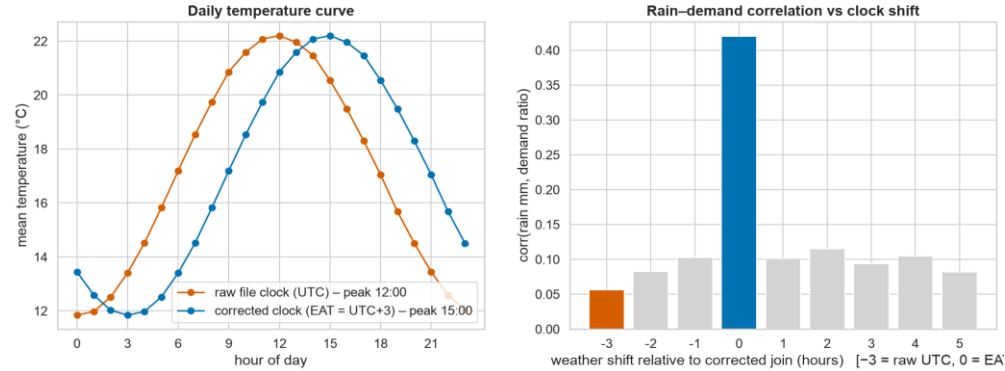
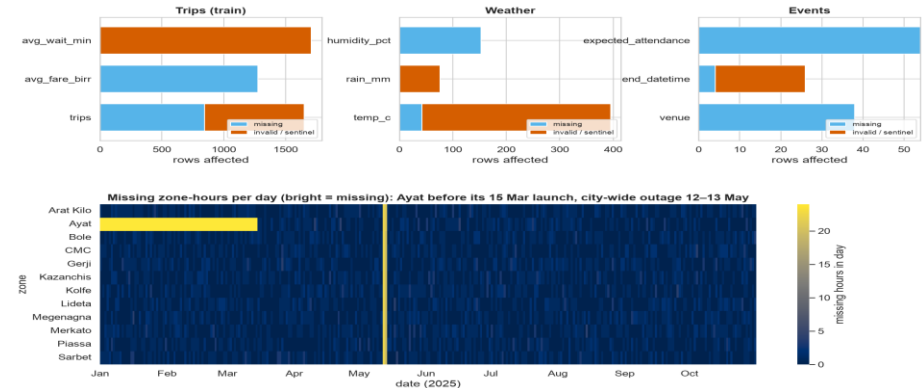
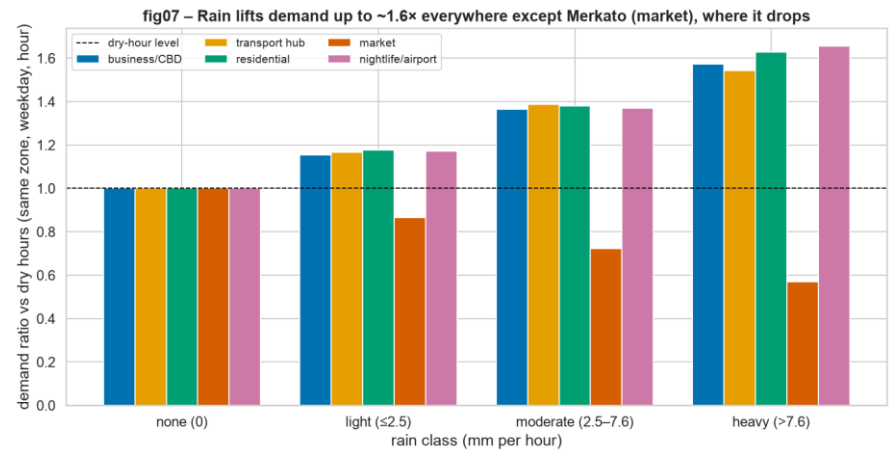


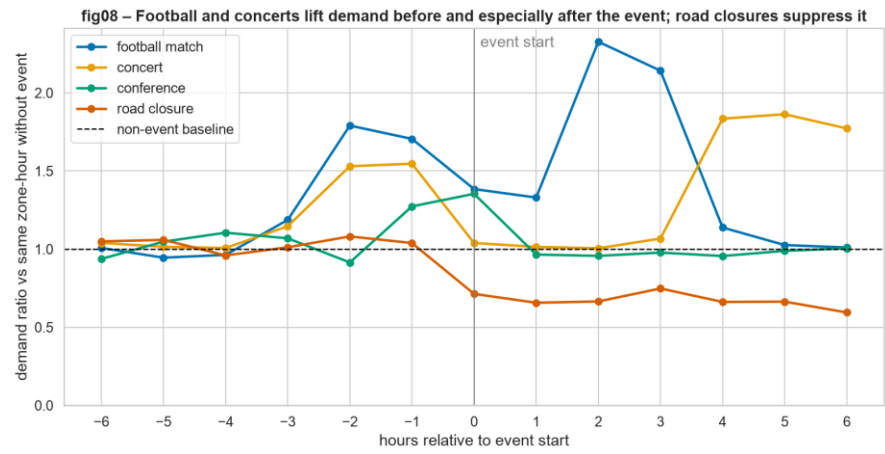
fig01 – Missing and invalid values in the raw tables, and gaps in the trip history



What the data says: rain and events move demand — but not everywhere



- **Rain:** +13% in light rain, +30% moderate, +46% heavy — sub-linear. But Merkato falls ~21%: an open-air market loses shoppers when it rains



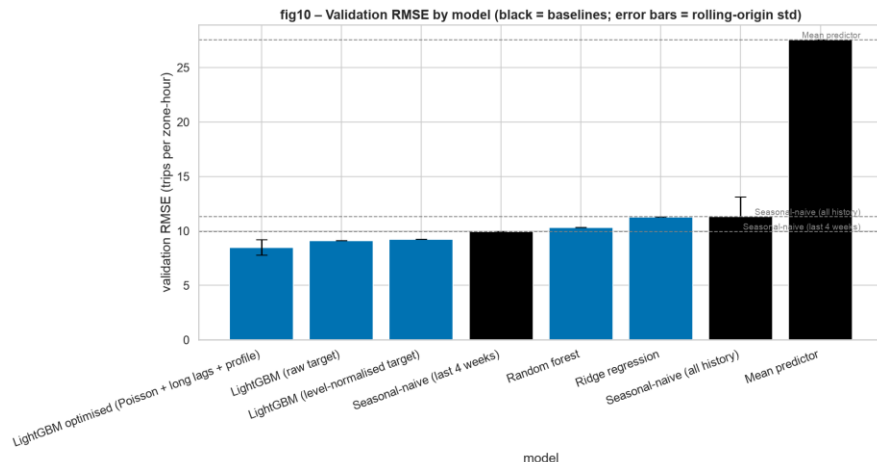
- **Events:** football lifts demand +75%, with the biggest surge in the 2 hours after the final whistle (+124%). Concerts +31%; road closures -24%. So event windows must extend past the end time

Modelling & evaluation: beat the baseline, honestly

Chronological validation: train before 18 Oct, test on 18–31 Oct, plus 4 rolling 14-day folds

Model	RMSE	MAE
Mean predictor	27.54	20.69
Seasonal-naive (zone×weekday×hour)	11.30	7.36
Ridge / random forest	11.29 / 10.31	7.55 / 6.76
LightGBM, tuned	9.03	6.06
LightGBM optimised (final)	8.47	5.85

- **Ablation:** calendar + lags 9.74 → + weather 9.21 → + events 9.64 → both 9.06: both joins help
- **Rolling origin:** 9.21 ± 0.71 vs 12.32 ± 1.83 for seasonal-naive, better in every fold
- **Leakage check:** adding active_drivers fakes an RMSE of 4.25, so it is excluded; only lags of 14 days or more are used
- **Optimisation:** Poisson loss, 2–5-week lags and an 8-week zone×weekday×hour profile take RMSE from 9.03 to 8.47



Error analysis, final score 8.47 RMSE, demo and next steps

- **Final score:** RMSE 8.47, MAE 5.85 trips per zone-hour ($\approx 17.5\%$ of demand). That is 25% better than seasonal-naïve and $\approx 4\text{--}5$ drivers per zone-hour
- **Where it misses:** busy zones in absolute terms (Bole 11.0, Megenagna 10.9, Merkato 10.5 vs Ayat 5.9) but relative error is flat (17–20%); rush hours; the tallest post-event surges; one-off holidays (Meskel)
- **Demo:** pick a zone and a date $\rightarrow$ 24-h forecast, peak hour, drivers needed, expected fares, and the weather/events looked up
 - run locally: `streamlit run app/app.py` (live demo on our laptop)
- **Next:** holiday- and event-specific multipliers, quantile forecasts for staffing, training with noisy weather forecasts, weekly retraining and hosting

fig11 – Forecast vs actual hourly trips, validation fortnight

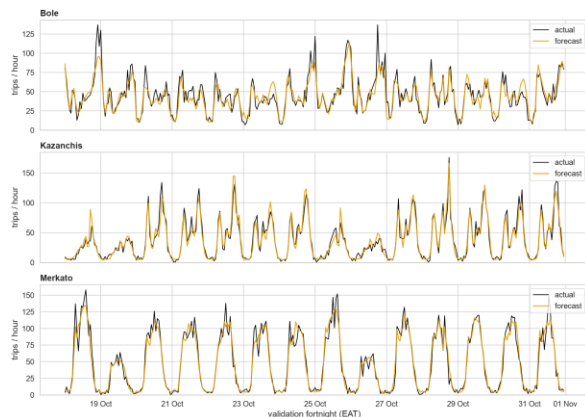
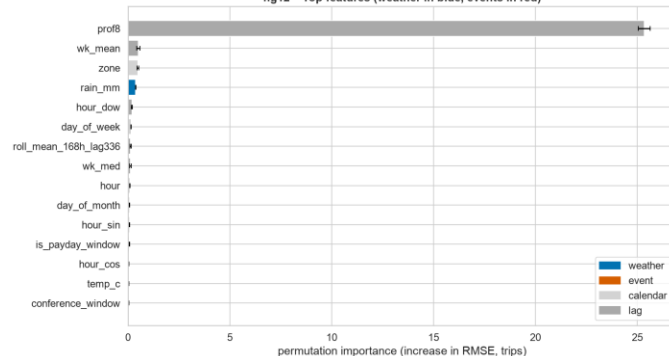


fig12 – Top features (weather in blue, events in red)



THANKS

